

Pekan 1 Praktikum Struktur Data



Praktikum

- 10 Pertemuan
- 1 pertemuan 1jam 40 menit sampai 2 jam
- Setiap mahasiswa diberikan 1 asisten

Penilaian

- Tugas Praktikum: 20% (H+2 Praktikum)
- Laporan Praktikum: 10% (H+2 Praktikum)
- Kelengkapan kode program di github: 20 % (H praktikum max jam 18.00)
- Ujian Akhir Praktikum: 50 %

Silabus Praktikum:

- | | |
|--------------|---|
| 1. Pekan 1 | ADT |
| 2. Pekan 2 | ArrayList |
| 3. Pekan 3 | Stack |
| 4. Pekan 4 | Queue |
| 5. Pekan 5 | Single Linked List |
| 6. Pekan 6 | Double Linked List |
| 7. Pekan 7 | Sorting 1 (Insertion, selection, Buble) |
| 8. Pekan 8 | Sorting 2 (shell, quick, merge) |
| 9. Pekan 9 | Searching (BFS dan DFS) |
| 10. Pekan 10 | Ujian |

Asisten M. Galid
github: **unVeroz007**

Kelas A

Asisten Sherly Sukmadira
Github: **Sherlysukmadira**

1	2411532007	NDARU PRADANA GATOT SUSENO
2	2511531019	KEVIN MAULANA SEMPRI
3	2511532001	AZ-ZAHRAND SOLICHUL TAJUSSALATHIN
4	2511532003	MUHAMMAD YUSUF INSY
5	2511532009	MUHAMAD FAJRI AULIA
6	2511532011	MARCELLO BAYU DENAR WIDJAKSONO
7	2511532019	TIARA AMALIA INSANI
8	2511532021	MUHAMMAD FAIZ AN- ANRI
9	2511532023	AMIRATUL FADHILAH
10	2511532025	NIDA TSABITA Ariba

11	2511532027	MUHAMMAD HANS NAFIS
12	2511532029	ABDUL KARIM ALGAZALI
13	2511533001	IHSANUL ZAKY EL- MUHAMMADY
14	2511533003	M. YAZEM AGVA ROIZ
15	2511533007	RIFKI MAULANA
16	2511533009	AFIF NAUFAL ZAHRAN
17	2511533015	SHOFWATUL HASANAH SARMA
18	2511533017	DIGO YUANDRA
19	2511533019	FILZI JELILA INDA ROBBANI
20	2511533027	NAIRA RAMADHANI HALIL

Asisten M. Galid
github: **unVeroz007**

Kelas B

Asisten Sherly Sukmadira
Github: **Sherlysukmadira**

1	2511531002	HAFIZH HABIBULLAH
2	2511531004	ALIF MUHAMMAD IHSAN
3	2511531006	LEXI MULIA YUNASPI
4	2511531008	MUHAMMAD IRFAN
5	2511531010	MUHAMMAD FHAREL
6	2511531012	MUHAMMAD AUFA RAFIKI
7	2511531014	NAYSHA APRILIA RIZKI
8	2511531020	DINDA AMELIA
9	2511532002	HALIYA ISMA HUSNA PUTRI AHMADI
10	2511532004	FAWWAZ KHALID
11	2511532008	RAHMAT ANANDA NAZAR
12	2511532010	HABIB AL FARUQ

13	2511532014	HAYKAL MUHAMMAD DAHNIL
14	2511532016	MUHAMMAD YASIN HABIBURRAHMAN
15	2511532020	ALHAMD RIDHO PUTRA
16	2511532022	THAARIQ SALAM
17	2511532026	ARYAHIYAHUL FIKRA
18	2511532028	RAZIF AL FARISI
19	2511533002	RIFQI ADITYA
20	2511533004	SILVA NURIFVAN
21	2511533006	ZAKI MISUCHA ANDEFI
22	2511533008	GHINADA FATHANAWAFA ALGMA
23	2511533014	GINA RAMADHANI
24	2511533028	ARDRA ATHAILLA FIRSY

Asisten M. Galid
github: **unVeroz007**

1	2511531001	DZHILLAN DZHALILA
2	2511531003	NABILA KHAIRUNNISA
3	2511531005	AFIF RAHMAN SALEH
4	2511531007	KHAULA LATHIFA FIRDAUSYI
5	2511531009	HAMDI SIDQI ALIFI
6	2511531011	ALIIFAH FELDA MUFARRIHATI SALWAA
7	2511531013	NOFRI ILHAM
8	2511531015	FAHRI SYAH PUTRA RAMADHAN
9	2511531017	ENDY PARDILIAN
10	2511532005	KHAIRUNNISA. M
11	2511532007	RAYYA SYAQINAH PUTRI HASIBUAN

Kelas C

Asisten Sherly Sukmadira
Github: **Sherlysukmadira**

12	2511532015	KHAIRUN NISA
13	2511532017	MUHAMMAD ARIF
14	2511533005	NAUFAL ARKAN OCTYANDI
15	2511533011	NABIL FIKRI
16	2511533013	ANNAFI AL GHIFARI
17	2511533021	ABDUL JABBAR
18	2511533023	M.FAJAR FADHILUL ZIKRI
19	2511533025	DIKA GIOWANDA
20	2511533029	SOFIAN ARBAI
21	2511533031	ALIYATAR RAFI AHMAD
22	2511537001	MUHAMMAD FATHAN EDLIN

Asisten M. Galid
github: **unVeroz007**

Kelas D

Asisten Sherly Sukmadira
Github: **Sherlysukmadira**

1	2311533016	RAFKI AHMAD PAGAMANDA
2	2511531016	KINAYA NOVRYA MANDA
3	2511531018	SYASYA HALWA GAZWANI
4	2511532006	RIZKY FADHLURRAHMAN ALMATRIA
5	2511532012	UMAR ABDUL AZIZ
6	2511532024	ANNISA LAYLI RAMADHANI
7	2511533010	HESKHIEL SURANTA HUTAUROK
8	2511533012	RAFIKHUL RAMADHAN

9	2511533016	MUHAMMAD FEDORA ARGADYAKSA
10	2511533018	MUHAMMAD ALTHAF MULYA
11	2511533020	DENNIS SHAUQI AKBAR
12	2511533024	RIFAL MAULANA OSKAR
13	2511533026	FAIZ FIKRI SATRIA
14	2511533030	AISYAH NAJMIATUL FAUZIAH
15	2511538002	JOSE MONIZ DE ARAUJO

Pendahuluan

- Buat repositori di github dengan nama: PrakSD2026_kelas_NIM
- Contoh: PrakSD2026_A_2511531000
- Collab akun github: informatika-fti dan akun asistennya

Praktikum Pekan 1

- Buat Project baru di eclipse dengan nama: PrakSD2026_kelas_NIM

Contoh: PrakSD2026_A_2511531000

- Setelah project berhasil dibuat pada folder src buat package: pekan1_nim
- Buatlah kode program berikut di dalam package pekan1_nim
- Setiap nama file tambahkan nim dibelakangnya:
- Contoh file: Jam.java menjadi Jam_2511531000.java

```

2
3 public final class Jam {
4     private int hh; // 0..23
5     private int mm; // 0..59
6     private int ss; // 0..59
7
8     // --- Validator ---
9 ⊖ public static boolean isValid(int h, int m, int s) {
10         return (0 <= h && h <= 23) && (0 <= m && m <= 59) && (0 <= s && s <= 59);
11     }
12
13     // --- Konstruktor/Kreator ---
14 ⊖ public Jam(int h, int m, int s) {
15         this.hh = h; this.mm = m; this.ss = s;
16     }
17
18     // --- Selektor ---
19     public int getHour() { return hh; }
20     public int getMinute() { return mm; }
21     public int getSecond() { return ss; }
22
23     // --- Mutator ---
24     public void setHour(int h) { this.hh = h; }
25     public void setMinute(int m) { this.mm = m; }
26     public void setSecond(int s) { this.ss = s; }
27

```

```

// --- Konversi ---
public int toSeconds() { return hh * 3600 + mm * 60 + ss; }
public static Jam fromSeconds(int total) {
    if (total < 0) throw new IllegalArgumentException("detik negatif");
    total %= 24 * 3600; // wrap 24 jam
    int h = total / 3600; total %= 3600;
    int m = total / 60; int s = total % 60;
    return new Jam(h, m, s);
}

// --- Relasional ---
public int compareTo(Jam other) { return Integer.compare(this.toSeconds(), other.toSeconds()); }
public boolean equals(Object o) {
    if (!(o instanceof Jam j)) return false;
    return hh == j.hh && mm == j.mm && ss == j.ss;
}
public int hashCode() { return java.util.Objects.hash(hh, mm, ss); }

// --- Aritmatika ---
public Jam plus(Jam other) { return fromSeconds(this.toSeconds() + other.toSeconds()); }
public Jam minus(Jam other) { return fromSeconds(Math.floorMod(this.toSeconds() - other.toSeconds(), 24*3600)); }
public Jam nextSecond() { return fromSeconds(this.toSeconds() + 1); }
public Jam nextNSeconds(int n) { return fromSeconds(this.toSeconds() + Math.max(0, n)); }
public Jam prevSecond() { return fromSeconds(Math.floorMod(this.toSeconds() - 1, 24*3600)); }
public Jam prevNSeconds(int n) { return fromSeconds(Math.floorMod(this.toSeconds() - Math.max(0, n), 24*3600)); }

// --- Durasi (detik) -- bisa negatif jika this > other? sesuai spesifikasi durasi(JAw, JAkh)
public static int durasiDetik(Jam jaw, Jam jakh) { return jakh.toSeconds() - jaw.toSeconds(); }
//overriding ke string
public String toString() { return String.format("%02d:%02d:%02d", hh, mm, ss); }
}

```

```

2
3 public class JamDriver {
4     public static void main(String[] args) {
5         Jam a = new Jam(23, 59, 50);
6         Jam b = new Jam(0, 0, 15);
7         System.out.println("a          = " + a);
8         System.out.println("b          = " + b);
9         System.out.println("a+b        = " + a.plus(b));
10        System.out.println("next 20s = " + a.nextNSeconds(20));
11        System.out.println("durasi(a,b) = " + Jam.durasiDetik(a, b));
12        System.out.println("a.compareTo(b) = " + a.compareTo(b));
13    }
14 }

```

Console X

<terminated> JamDriver [Java Applic

```

a          = 23:59:50
b          = 00:00:15
a+b        = 00:00:05
next 20s   = 00:00:10
durasi(a,b) = -86375
a.compareTo(b) = 1

```

```
3 import java.util.Scanner;
4 public class JamDriver2 {
5     public static void main(String[] args) {
6         Scanner input = new Scanner(System.in);
7         System.out.println("=== Program Driver Objek Jam ===");
8         // 1. Input Jam Pertama
9         System.out.println("\n--- Input Jam 1 ---");
10        Jam j1 = buatJamDariInput(input);
11
12        // 2. Input Jam Kedua
13        System.out.println("\n--- Input Jam 2 ---");
14        Jam j2 = buatJamDariInput(input);
15
16        // 3. Menampilkan Data
17        System.out.println("\n--- Hasil Operasi ---");
18        System.out.println("Jam 1 (String)           : " + j1.toString());
19        System.out.println("Jam 2 (String)           : " + j2.toString());
20        System.out.println("Jam 1 dalam detik      : " + j1.toSeconds());
21        System.out.println("Jam 2 dalam detik      : " + j2.toSeconds());
22    }
```

```
// 4. Operasi Relasional (Perbandingan)
```

```
int perbandingan = j1.compareTo(j2);
```

```
if (perbandingan > 0) {
```

```
    System.out.println("Status
```

```
: Jam 1 lebih lambat (setelah) Jam 2");
```

```
} else if (perbandingan < 0) {
```

```
    System.out.println("Status
```

```
: Jam 1 lebih awal (sebelum) Jam 2");
```

```
} else {
```

```
    System.out.println("Status
```

```
: Jam 1 dan Jam 2 sama persis");
```

```
}
```

```
// 5. Operasi Aritmatika
```

```
System.out.println("Durasi (J1 ke J2)      : " + Jam.durasiDetik(j1, j2) + " detik");
```

```
Jam jNext = j1.nextSecond();
```

```
System.out.println("Jam 1 Detik Berikutnya: " + jNext);
```

```
Jam jPrev = j1.prevSecond();
```

```
System.out.println("Jam 1 Detik Sebelumnya: " + jPrev);
```

```
// 6. Operasi Penjumlahan Jam
```

```
Jam jHasilPlus = j1.plus(j2);
```

```
System.out.println("Hasil J1 + J2          : " + jHasilPlus);
```

```
input.close();
```

```
System.out.println("\nProgram Selesai.");
```

```
}
```



```

}
/**
 * Prosedur pembantu untuk melakukan input dan validasi secara berulang
 * sampai user memasukkan angka yang benar (0..23, 0..59).
 */
private static Jam buatJamDariInput(Scanner sc) {
    int h, m, s;
    while (true) {
        System.out.print("Masukkan Jam (0-23)    : ");
        h = sc.nextInt();
        System.out.print("Masukkan Menit (0-59) : ");
        m = sc.nextInt();
        System.out.print("Masukkan Detik (0-59) : ");
        s = sc.nextInt();

        // Memanggil method static isValid dari kelas Jam
        if (Jam.isValid(h, m, s)) {
            return new Jam(h, m, s);
        } else {
            System.out.println("[Error] Input tidak valid! Silakan ulangi.\n");
        }
    }
}
}
}

```